

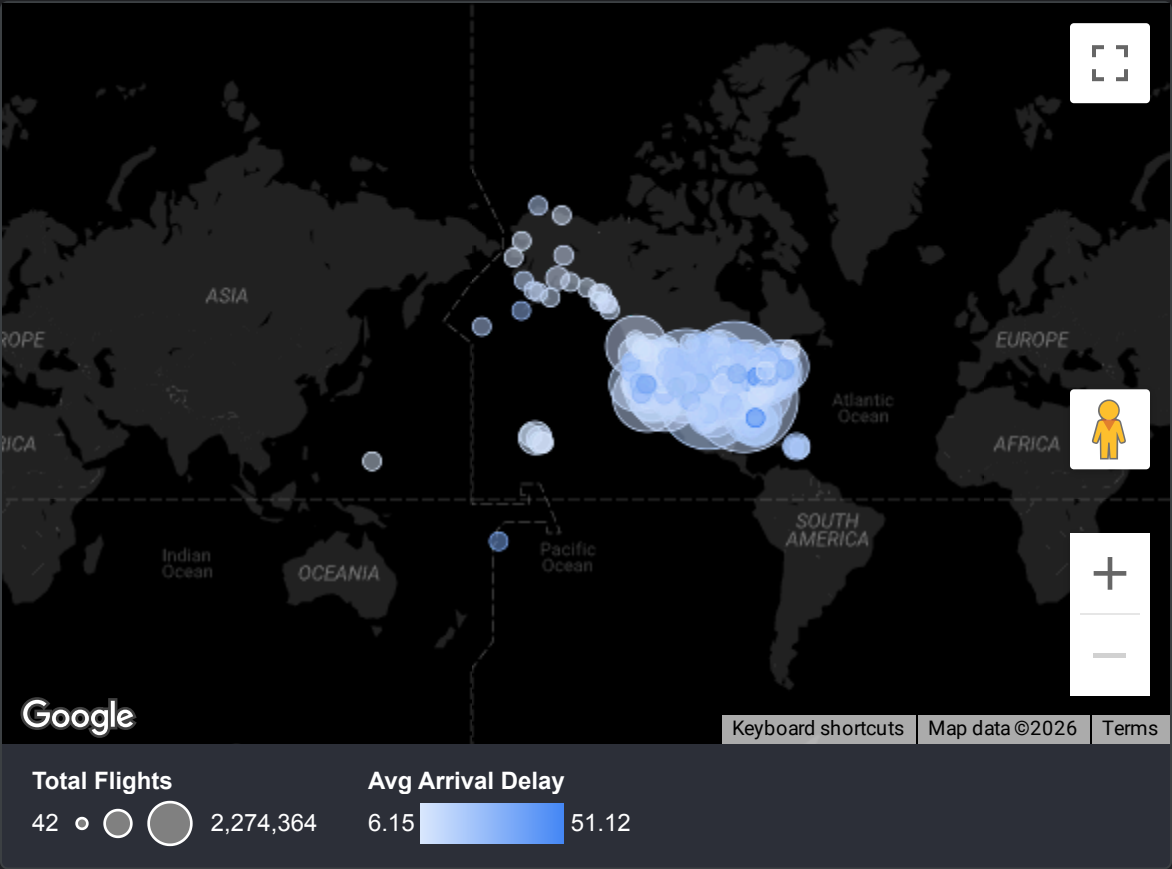
U.S. Airline Flight Delay Analysis Pipeline

Jan 1, 2019 - Dec 31, 2025

Airline

Airport Code

Airport delay heatmap

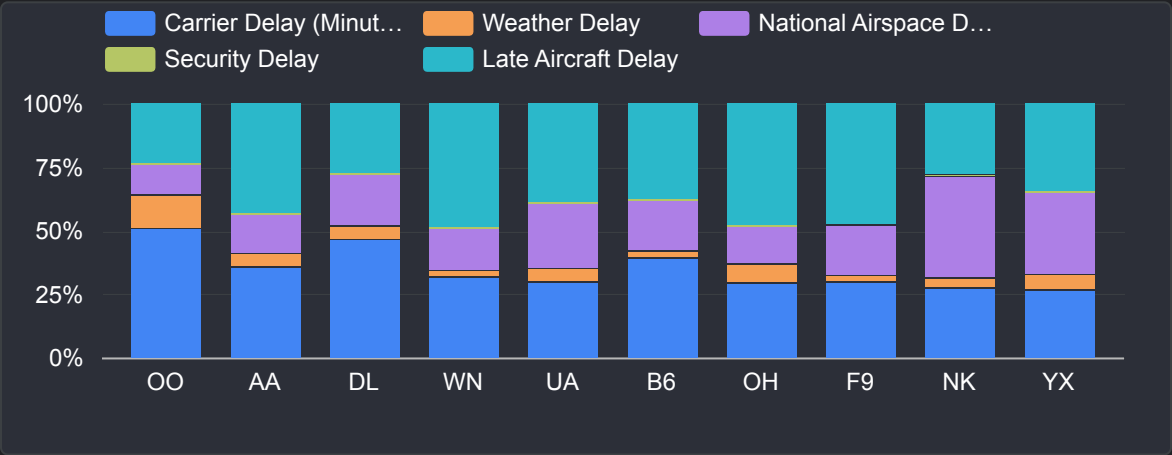


Airline KPI table

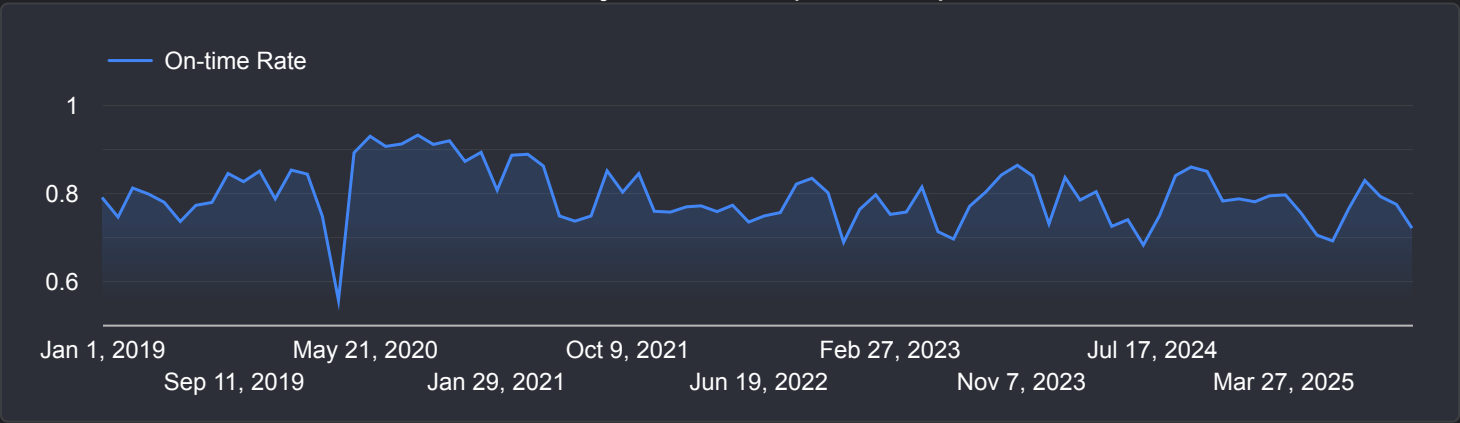
	Airline	Avg Arrival Delay	Flights	On-time %	Cancellation %
1.	WN	11.44	8,946,780	78.16	2.41
2.	DL	11.72	6,234,749	83.28	1.26
3.	AA	18.2	6,025,616	76.14	2.32
4.	OO	14.9	5,180,101	81.09	1.89
5.	UA	14.87	4,295,371	78.91	1.75

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Delay cause breakdown by airline



Monthly on-time rate (2019–2025).



Weather vs delay rate

Visibility / Delay Rate			
Wind Sp...	poor (<3 mi)	reduced (3-10 mi)	clear (>=10 mi)
00-05 kt	24.47%	21.09%	18.89%
05-10 kt	30.17%	22.78%	18.54%
10-20 kt	39.57%	24.46%	18.36%
20-30 kt	31.26%	27.7%	32.57%
30+ kt	5.3%	15.34%	14.78%

BigQuery ML Classifiers

Model Comparison

	model_type	roc_auc	accuracy	precision ▾	recall	f1	log_loss
1.	boosted_tree	0.65	0.59	0.32	0.66	0.43	0.68
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Model Type: boosted_tree

(1) ▾

Predicted / Flights		
Actual	predicted_on_time	predicted_delayed
0	3,035,278	2,328,827
1	559,752	1,077,762

ROC-AUC

0.65

Accuracy

0.59

Precision

0.32

Recall

0.66

F1

0.43

Log loss

0.68

